

Ding-Jiun (Willy) Huang

Email: dingjiun.huang@mail.utoronto.ca | (+1) 942-381-2491 | Homepage: <https://willydjhuang.github.io>

EDUCATION

University of Toronto (UofT)

Toronto, ON

PhD in Computer Science

Advisor: Prof. David Lindell

Carnegie Mellon University (CMU)

Pittsburgh, PA

M.S. in Computer Vision (GPA 3.95/4.0)

Dec. 2025

Advisor: Prof. Fernando De la Torre

National Taiwan University (NTU)

Taipei, Taiwan

B.S. in Computer Science and Information Engineering (GPA 4.17/4.3)

Jun. 2023

RESEARCH INTERESTS

3D Reconstruction and Generation, 3D Garment Synthesis, Human Avatar

PUBLICATIONS

OmniFabric: Coherent UV Space Texture Synthesis for 3D Garment Reconstruction

Ding-Jiun Huang, Yuanhao Wang, Cheng Zhang, Hugo Bertiche, Alexandru-Eugen Ichim, Thabo Beeler, Fernando De la Torre

SIGGRAPH Asia, 2026. [[paper](#)]

From Blurry to Believable: Enhancing Low-quality Talking Heads with 3D Generative Priors

Ding-Jiun Huang, Yuanhao Wang, Shao-Ji Yuan, Albert Mosella-Montoro, Francisco Vicente Carrasco, Cheng Zhang, Fernando de la Torre

3DV, 2026. [[project page](#)]

ASSR-NeRF: Arbitrary-Scale Super-Resolution on Voxel Grid for High-Quality Radiance Fields Reconstruction

Ding-Jiun Huang, Zi-Ting Chou, Yu-Chiang Frank Wang, Cheng Sun

arxiv, 2024. [[project page](#)]

Consensus-Based Fault-Tolerant Platooning for Connected and Autonomous Vehicles

Tzu-Yen Tseng, Ding-Jiun Huang, Jia-You Lin, Po-Jui Chang, Chung-Wei Lin, Changliu Liu

IEEE Symposium on Intelligent Vehicle, 2023. [[paper](#)]

SB-VQA: A Stack-Based Video Quality Assessment Framework for Video Enhancement

Ding-Jiun Huang, Yu-Ting Kao, Tieh-Hung Chuang, YaChun Tsai, Jing-Kai Lou, and Shuen-Huei Guan

CVPR NTIRE, 2023. [[paper](#)]

RESEARCH EXPERIENCE

Carnegie Mellon University, Human Sensing Laboratory

Pittsburgh, PA

Graduate Researcher, Sponsored Collaboration with Google Research

08/2024~Present

Advisor: Prof. Fernando De la Torre, Prof. Cheng Zhang, Dr. Thabo Beeler

- Enhance 3D head avatars created from low-quality head motion captures by utilizing 2D generative priors
- Propose a dynamics-aware 3D GAN inversion, utilizing both multi-view and multi-expression facial captures to synthesize 3D head avatar with high-fidelity geometry and detailed textures
- Propose a data-driven approach to generate a simulation-ready textured garment from a single clothing image through 3D-aware texture generation with diffusion models on predicted 2D sewing patterns

National Taiwan University, Vision and Learning Laboratory

Taipei, Taiwan

Graduate Researcher, Collaboration with NVIDIA Research Taiwan

08/2023~06/2024

Advisor: Prof. Yu-Chiang Frank Wang, Dr. Cheng Sun

- Enhance general-scene NeRF representations built from low-resolution multi-view captures, outperforming Zip-NeRF in PSNR/SSIM and achieving superior perceptual quality
- Propose a voxel-based 3D super-resolution method to enhance NeRF scenes with only a feed-forward

National Taiwan University, Cyber-Physical Systems Laboratory

Undergraduate Researcher

Taipei, Taiwan

07/2022~06/2023

Advisor: Prof. Chung-Wei Lin

- Design a fault-proof communication protocol to enhance stability of autonomous vehicles platooning
- Recover platooning affected by malicious attack with 43% reduction in system settling time over SOTA method

Arizona State University, Make Programming Simple Laboratory

Undergraduate Researcher, Remote collaboration

Tempe, AZ

07/2022~06/2023

- Explore motion planning algorithms for autonomous vehicles that fits b-spline curves to predict waypoints of vehicles, effectively avoiding possible collisions in a multi-lane highway scenario

WORK EXPERIENCE

KKCompany, Advanced Research Center

Research Intern

Taipei, Taiwan

07/2022~06/2023

- Build an automatic pipeline to enhance low-quality film content using video super-resolution and interpolation
- Explore real-time video super-resolution in an online streaming scenario
- Propose SB-VQA, a quality assessment method for videos processed by deep learning enhancement methods

TECHNICAL SKILLS

C/C++, CUDA, Pytorch, Python, Tensorflow, JAX, OpenCV, MATLAB, Unity, Blender

RELEVANT COURSEWORK

Carnegie Mellon University: Advanced Computer Vision, Human Robot Interaction, Image Synthesis, Visual Learning and Recognition

National Taiwan University: Deep Learning for Computer Vision, Computer Graphics, Deep Learning for Human Language Processing, Computer Architecture, Algorithm Design and Analysis, Systems Programming, Network Administration and System Administration

SELECTED PROJECTS

Reinforcement Learning with In-Memory Computing

RL & System Research Project at NTU, Advisor: Prof. Tei-Wei Kuo

- Designed a ReRAM-based accelerator as well as a gate-level pipeline to improve the efficiency of reinforcement learning to mitigate the memory bottleneck problem with Von-Neumann architecture.
- Reduced the overhead to move heavy model weights between GPU and memory devices, the proposed method reaches an x2.1 speedup in training.
- Received *Most Promising Research Work by Appier, NTU CSIE Research Projects Exhibition 2021*

JetFinger: Devices for Immersive Gaming Experience

HCI Research Project at NTU, Advisor: Prof. Tei-Wei Kuo

- Aimed to create a VR sword fighting game with immersive gaming experience by producing realistic feedback on users' VR handheld.
- Designed an air-propulsion device attached to a VR handheld, creating a reaction force whenever the user hits something with a sword in the game.

NASA Monitoring System

DevOps Engineering Project at NTU

- Built a web service to monitor all the servers of the Network and System Administration (NASA) team at NTU CSIE, leveraging techniques including Prometheus, Grafana, LDAP, Proxy Configurations.
- Configured the Proxy, LDAP and IP settings among the machines of different groups in the NASA team.

TEACHING EXPERIENCE

Data Structure and Algorithm, course at NTU

Teaching Assistant

- Designed assignments of algorithms for more than 150+ students in this compulsory course for CSIE majors and taught students with advanced data structures and algorithms in scheduled TA hours.
- Received Outstanding Teaching Assistant Award by NTU CSIE.

Network Administration and System Administration, course at NTU

Teaching Assistant

- Taught students about applications related to network and system administration including firewall, DNS, web service, virtual machine management with hands-on implementations as TA.

NTU Shepherd Project

Computer Science Mentor

- In both online courses and in-person classes, I taught a total of 300 high school students topics including data structure and algorithm, web service, machine learning.
- Designed interactive learning materials such as online playgrounds for HTML, CSS, Javascript and image classification challenge with PyTorch.